

Jake Walklate

[0435 739 578](tel:0435739578) | jakew@jakewalklate.is-a.dev | [linkedin.com/in/jakewalklate](https://www.linkedin.com/in/jakewalklate) | www.jakewalklate.is-a.dev

CAREER SUMMARY

Penultimate dual-degree student seeking an engineering or research-focused role where I can utilise my core cross-disciplinary systems knowledge. Passionate about creating web and embedded software, prototyping physical hardware, and analysing biological data. Previous industry experience writing software code and a proven ability to manage and take ownership of technical decisions and demonstrate a collaborative nature.

I have a rare and technically challenging interdisciplinary skill set, consisting of medical science and computer related engineering. The added value is in my ability to fluidly translate and collaborate across and between independent and siloed teams needing support, contribute research principles, build end-to-end data processing systems, and providing engineering design and development.

EDUCATION

University of Technology Sydney

Ultimo, NSW

Bachelor of Engineering (Honours), in Artificial Intelligence & Mechatronics

2021 – Jun 2028 (exp.)

- * Distinction in Mathematical Modelling
- * High Distinction in Computer Programming
- * High Distinction in Mechatronic Engineering
- * Distinction in Engineering Projects

Bachelor of Medical Science, in Pathology

2020 – Jun 2028 (exp.)

- * Distinction in 3D Biomedical Polymers

TAFE NSW

St Leonards, NSW

Certificate IV of Information Technology, in Cyber Security

Graduated December 2020

Certificate IV of Information Technology, in Software Development

Graduated December 2019

EXPERIENCE

Founding Engineer

Mar 2021 – Jan 2022

Coreable

Sydney (Remote)

- Led the backend and co-developed a full-stack hiring and soft-skills assessment platform used by hundreds of university students in a successful MVP trial.
- Designed and implemented the backend (GraphQL Apollo Server, Node.js, Express.js, MySQL, Sequelize) with authentication (hashing with bcrypt and JWT cookies) for API security, object-oriented data models, and non-blocking async architecture for low latency and high-speed server processing.
- Contributed to front-end React.js component development with TypeScript and SCSS (HTML, CSS, JavaScript) using stateful component design principles, react hooks and secure code practices.
- Collaborated with a small team of transdisciplinary UI/UX designers, stakeholders and entrepreneurs, including academic mentors and facilitator/investors on the team which was winners of the 2018 UTS Startups hatchery award.
- Worked in an Agile environment: participated in sprints, code reviews, GitHub workflow automation, Docker test environments, and Google Cloud Platform (GCP) based deployment via CI/CD pipelines.

Full-Stack Software Developer Intern

Jun 2019 – Feb 2020

ClockOn

West Gosford, NSW

- Responsible for the technical execution of a major enterprise application rewrite from a legacy Delphi windows application to a modern RESTful HTML, C#, SQL, CSS (Bootstrap) & JavaScript web stack by following web accessibility guidelines and modern UX recommendations.
- Developed and implemented engineering solutions using third-party UI libraries (Vue.JS), with focus on cross-browser support and accessibility, collaborating with project stakeholders in an Agile methodology.

PROJECTS

Custom 3D Printer Upgrade | C++, Python

2026

- Engineered a dual-zone thermal regulation system for an Anycubic Kobra S1 3D printer by utilising a parallel 24V heating array, high-pressure CPAP centrifugal blower and sound-proofed enclosure to improve vertical polymer adhesion and maintain usability.

- Developed non-blocking and self-configuring C++ Arduino firmware on an ESP32 MCU featuring automatic Wi-Fi reconnection, mDNS service broadcast and discovery, and a RESTful HTTP endpoint for remote telemetry and automated heating control.
- Repaired a broken camera egress function by reverse engineering compiled C binaries on a Rockchip SoC using Ghidra and reusing shared object libraries in a custom cloud pusher script configured via SSH with Python scripts running as looping Linux cron jobs for continuous hardware monitoring.
- Ported DOOM to the onboard touch display using fbDOOM compiled on Docker for ARMv7.

Multilayer Perceptron Classifier | Python2026

- Created an Artificial Neural Network (MLP) from scratch with a bipolar activation function, gradient descent backpropagation and manual momentum optimisation using only NumPy (and no further libraries or plugins) for classification of digit inputs in Python.

International Space Station Tracker | C++, Arduino2025

- Modelled a 3D printable robot arm for the purpose of housing an Arduino based IoT project using Autodesk Fusion (360) and selected appropriate polymer materials (such as PA-6/12) and greases (PTFE) for bearings for lowest friction and wear, and ASA for tough and UV resistant structural bodies.
- Called external HTTP APIs to find ISS TLE data, having hard-coded backups in case of network failure, then gathered positional data by using GPS, then calculating the resulting direction with an external SGP4 library, control to create a physical pointer that follows the ISS across the sky. When the station passes overhead, the device rotates to track its position in real-time.

Blood Cell Neural Network Classifier | TensorFlow2024

- Used a Kaggle dataset to build a convolutional neural network, and a multiplayer perceptron classifier with TensorFlow's sequential API and one-hot encoding to automate and distinguish different types of blood cells under microscopy.

Computational Simulation of Human Genetic Inheritance | Rust2022

- Created computer simulation software to model the heritability of genetic traits involved in partner attraction in humans to determine the percent wise contribution of social and environmental factors to sexual behaviour.
- Used Rust as the language to avoid garbage collectors' memory overhead to model the earth's population to 1.3 billion individuals on 16GB of RAM, determining an equilibrium of genetic inheritance to contribute to ongoing psycho-sociological research.

OTHER EXPERIENCE

Team Lead (Shift Supervisor) <i>Pattysmiths Burgers</i>	Sep 2024 – Current <i>Darlinghurst, NSW</i>
• Contribute to a fast-paced restaurant environment, cooking customer orders, and cleaning. • Voted favourite team member by management and peers.	
Fiber & Ethernet Network Cabler <i>Mirait Technologies</i>	2015 <i>Sydney, NSW</i>
• Spliced fiber optic cables and connected houses to the National Broadband Network rollout. • Debugged network faults using testing tools and repaired conduit and terminations.	

VOLUNTEER WORK

Member & Contributor <i>Tech Shed Connections</i>	May 2026 – Current <i>Canada Bay Library</i>
• Volunteer for the Australian Men's Shed Association, participating in technology conferences, membership meetings, and educational sessions (with verified Working with Children Check). • Teach Linux skills, repair vintage tech hardware, contribute to woodworking, and configuring software with vulnerable and retired engineers and technical workers. • Build 3D printed fighting robots with ESP32 microcontrollers for the purpose of running regular educational sessions for local school children to teach them about STEM projects.	

MEMBERSHIPS

Engineers Australia | Student Associate
Google Developer Group Sydney | Member

REFERENCES

Available on request.